

CYBERSECURITY RESEARCH PROJECT

PHISHGUARD

A Real-Time Anti-Phishing Browser Extension

Developed by **Worship Muguti**

| THE PROBLEM

EXPLOITING TRUST

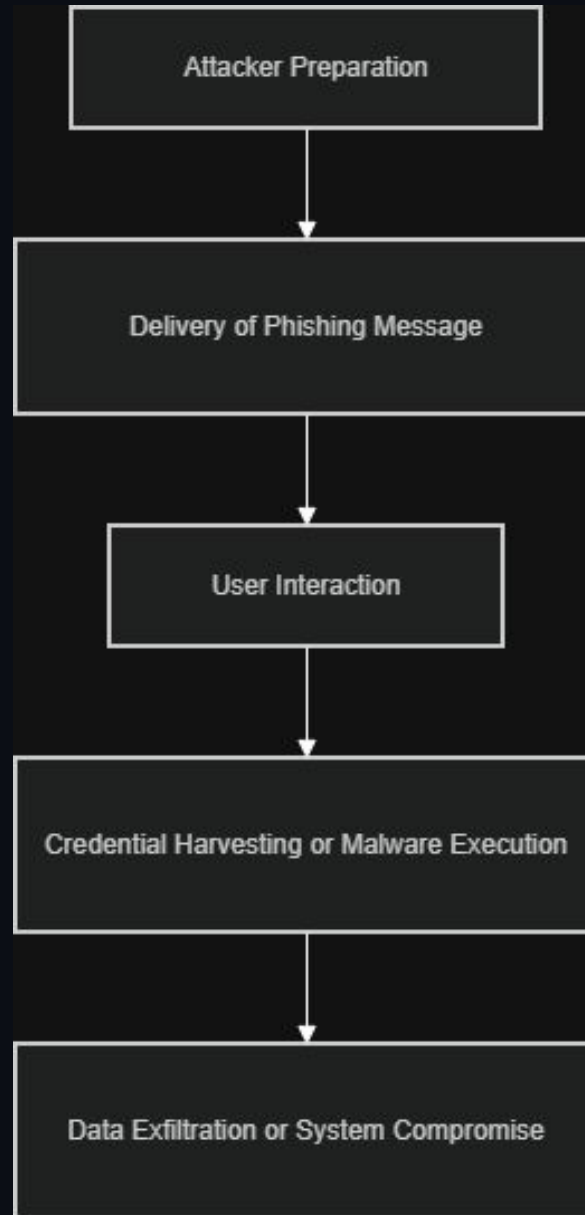
Phishing attacks remain the most pervasive cybersecurity threat, tricking users into revealing passwords and financial data.

 Bypasses standard vigilance

 Malicious links via Email & Messaging

 Automation is now a necessity

| THE PROCESS



| REQUIREMENT ANALYSIS



USER NEEDS

Non-technical users require a seamless experience.
Protection must be invisible until a threat is detected,
using subtle alerts rather than intrusive popups.



SYSTEM NEEDS

Cross-browser compatibility (starting with Chrome).
Low resource footprint is critical to ensure zero
degradation of browsing speed.

| CURRENT MARKET GAPS

Existing Solution	Primary Method	The PhishGuard Advantage
Google Safe Browsing	Blacklist Only	Real-time ML Analysis for Zero-day threats
Norton Safe Web	Proprietary Lists	Open Source Transparency & Hourly Updates
Avast Security	Bulky Extension	Lightweight Client-side processing

| OPEN SOURCE INTELLIGENCE



PHISHTANK

Community-driven blacklist with hourly API updates. Core of our dual-layered lookup system.



OPENPHISH

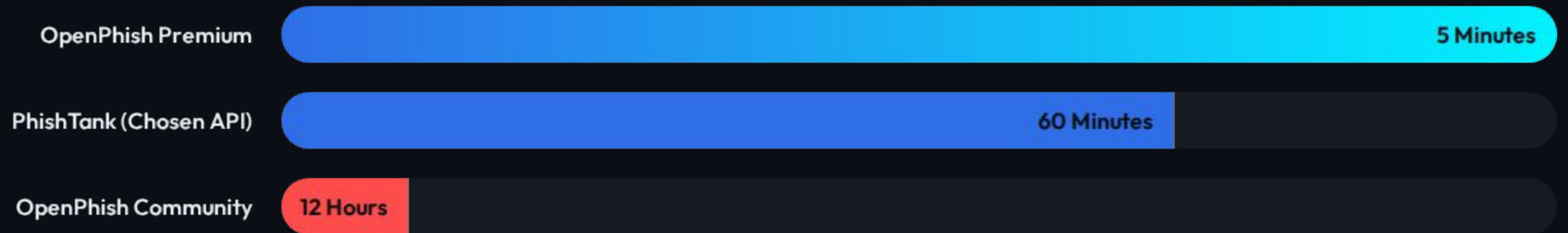
Specialized active feed. PhishGuard accounts for the 12-hour free-tier delay.



URLHAUS

Used to cross-reference malware distribution domains for multi-threat detection.

| DETECTION FRESHNESS



Update frequency is the #1 factor in catching "Short-lived" phishing pages.

| MACHINE LEARNING LAYER

CLASSIFICATION FEATURES

-  **URL Features:** Length, IP address presence, and suspicious character counts.
-  **Domain Data:** Age of domain, registration length, and keyword red-flags.
-  **Algorithms:** Random Forest and Decision Trees via Scikit-learn.

RISK ANALYSIS

The classifier calculates a risk score for any URL not found on established blacklists, identifying novel threats in milliseconds.

| STRATEGIC GOALS

98%

IF EVERYTHING WORKS CORRECTLY, IT SHOULD
HAVE A DETECTION ACCURACY TARGET OF 98%

PERFORMANCE STANDARDS

My focus is minimizing the **False Positive Rate** to ensure the extension remains helpful rather than intrusive. Success is measured by accuracy vs browser load impact.

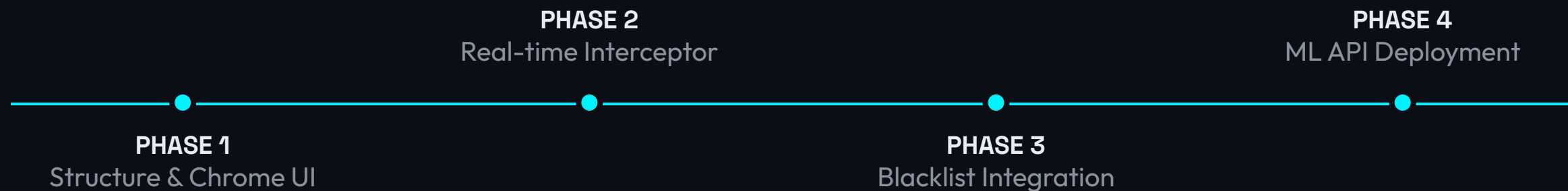
| TECH FEASIBILITY

PROVEN STACK

Leveraging well-documented APIs like Chrome Extensions and PhishTank. The ML component is built using industry-standard Python Scikit-learn.

The solution is financially viable, relying entirely on open-source tools and free API tiers, keeping development costs minimal.

| DEVELOPMENT ROADMAP



| DEMONSTRATION OF HOW THE EXTENTION SHOULD WORK

<https://youtu.be/sP9XmjJkEc?si=MOrXoyCeZM9924sS>

THE FUTURE OF PHISHGUARD

Expanding protection to Firefox and Edge, implementing real-time email content analysis, and launching a centralised enterprise dashboard for organisational threat monitoring.

QUESTIONS?

Secure your browsing experience with PhishGuard.

Worship Muguti | Project Lead
Real-Time Cybersecurity Solutions